

AFOSR/NASA ULI FAST Year 3 Review Meeting: Poster Session

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Predicting Uniform temperature Change



Predicting Uniform Temperature Change

- When a structure experiences uniform temperature change, the either contracts or expands.
- This contracting or expansion can be measure in the form of strain and this is defined as thermal strain
- The methodology is applied to the IC3X benchtop model



Predicting Uniform Temperature Change

$$f_{a \times n}$$

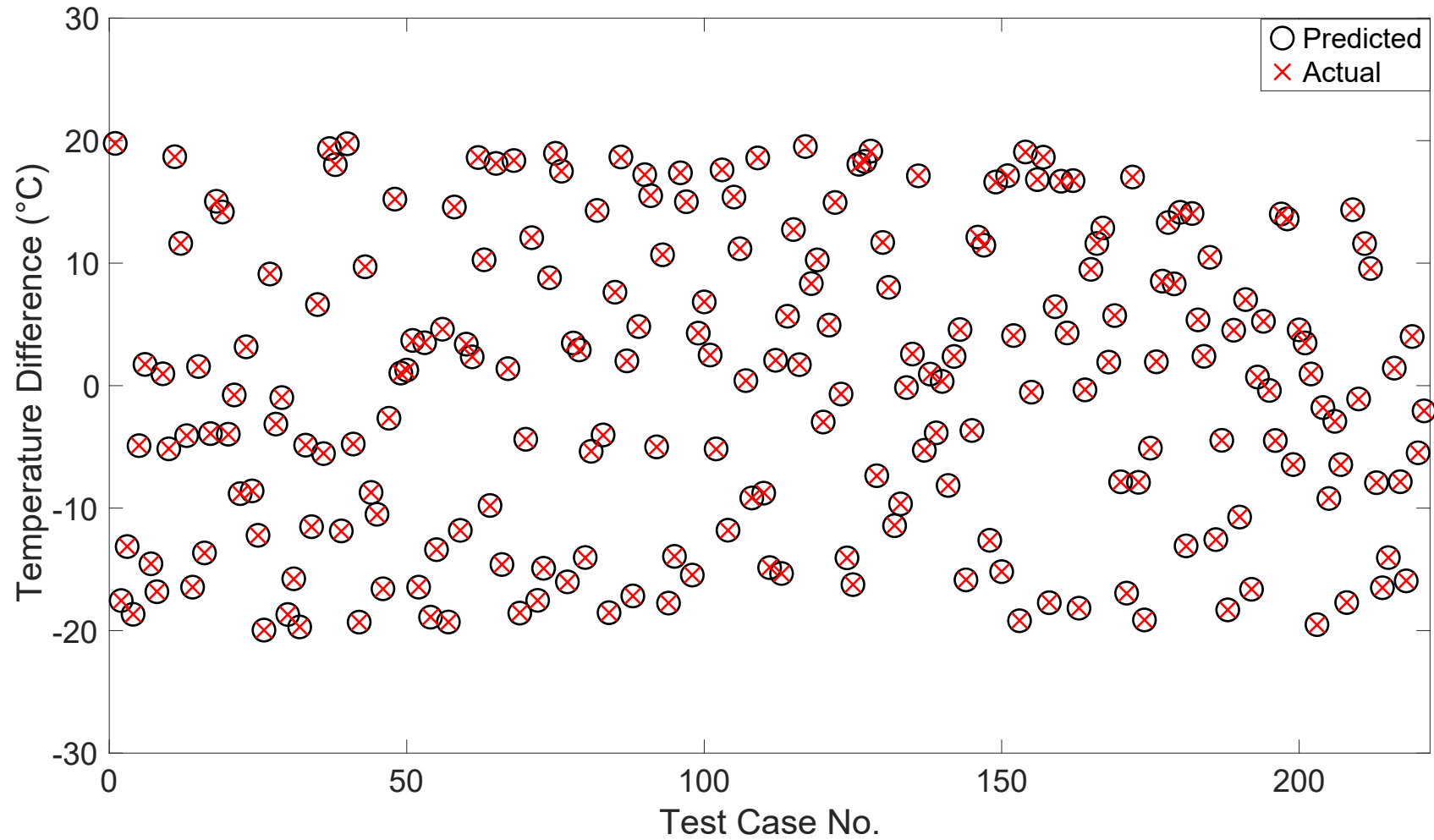
$$\Delta T_{1 \times n}$$

$$\{\epsilon_{32 \times t}\} = \sum_{n=1}^4 \frac{f_{n \times t} \cdot (l_f - l_i) \cdot y_i}{EI_i} + \alpha \Delta T_n \text{ for } l_i < l_f$$

$$[C_{i \times a}] = \{\epsilon_{i \times n}\} \{\Delta T_{1 \times n}, f_{a \times n}\}^T ((\Delta T_{1 \times n}, f_{a \times n}) \cdot (\Delta T_{1 \times n}, f_{a \times n})^T)^{-1}$$

$$\{\Delta T_p, f_p\} = (C^T \cdot C) \cdot C^T \{\epsilon\} = [T] \{\epsilon\}$$

Result: Predicting Uniform Temperature Change



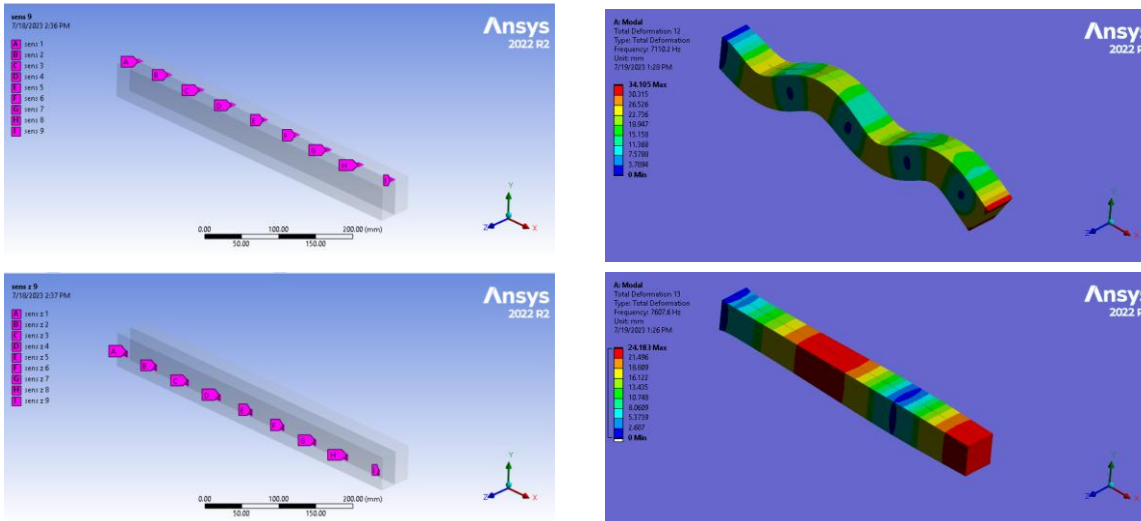
Strain to Deformation Method



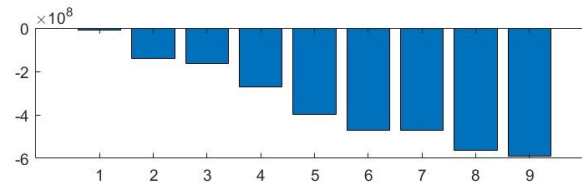
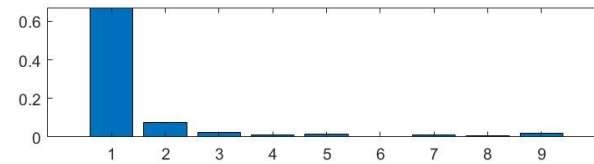
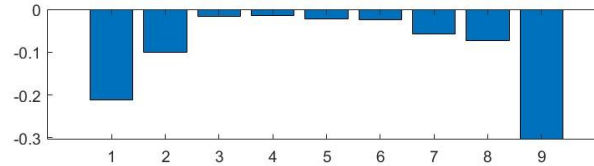
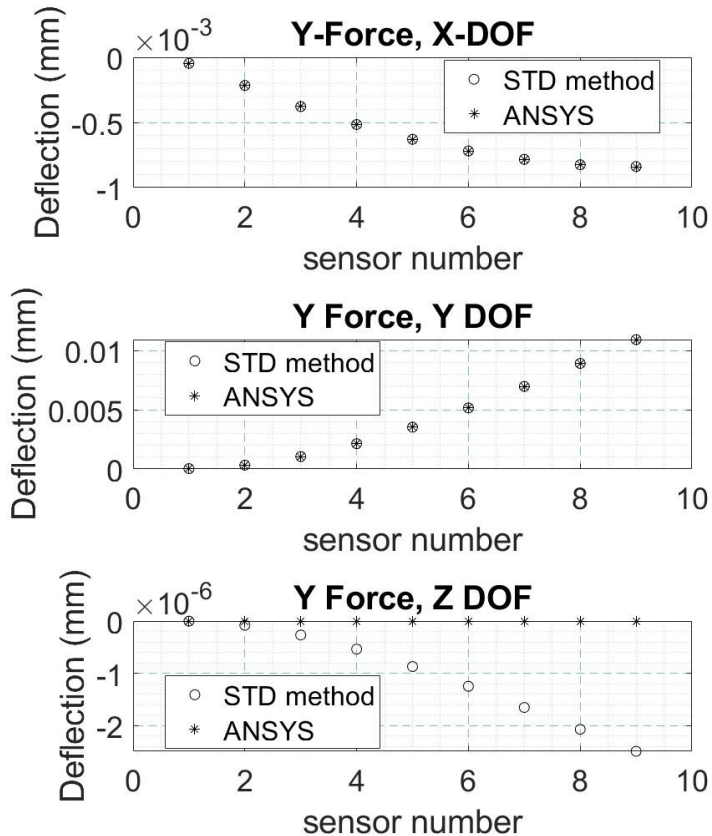
Strain to Deformation Method: Overview

$$\begin{aligned}\{u\} &= [\Phi]\{\xi\} \\ \{\epsilon\} &= \frac{\partial u}{\partial x} = \frac{\partial}{\partial x} [\Phi]\{\xi\} = [\Psi]\{\xi\} \\ \{\xi\} &= [[\Psi]^T [\Psi]]^{-1} [\Psi]^T \{\epsilon\} \\ \{u\} &= [\Phi] [[\Psi]^T [\Psi]]^{-1} [\Psi]^T \{\epsilon\}\end{aligned}$$

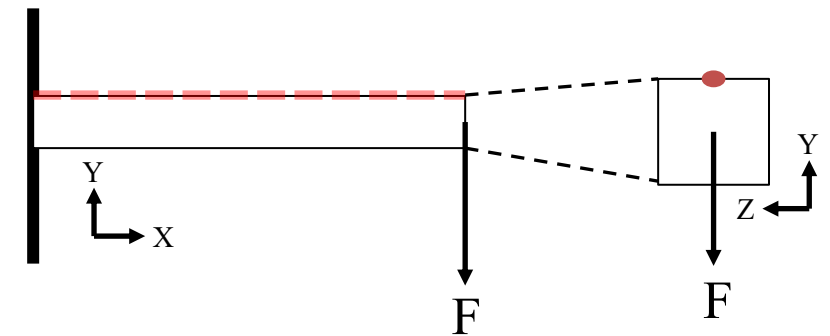
- From “Fiber-Optics-Based Aeroelastic Shape Sensing” by Freydin et al.
- In the equation, $\{u\}$ represents the displacement vector of the sensors, $[\Phi]$ represents the matrix of displacement modes, $\{\xi\}$ is the modal displacements, $\{\epsilon\}$ is the vector of strains, and $[\Psi]$ is the matrix of strain modes.
- Normal strain and deformation mode shapes generated from ANSYS Modal Analysis
- Modes chosen that interact with each respective degree of freedom – modal participation factor
- Normal strain readings in each displacement axis for respective tip forces collected from ANSYS Static Structural Analysis, collecting average values at each “virtual sensor”
- Coefficient matrix constructed from mode shapes, then used to test respective strains and forces from simulation



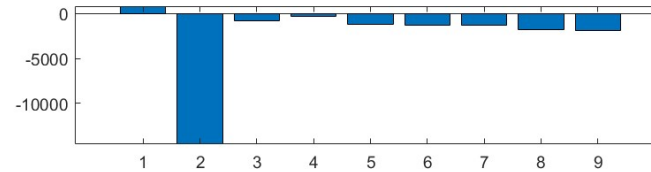
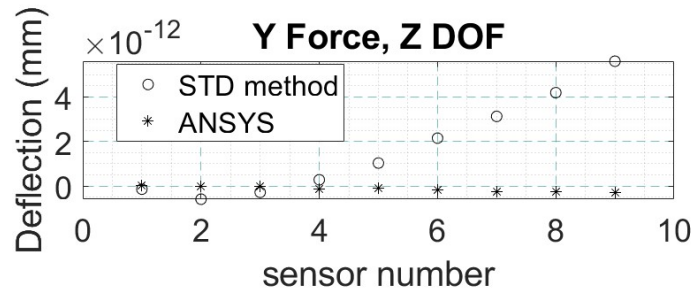
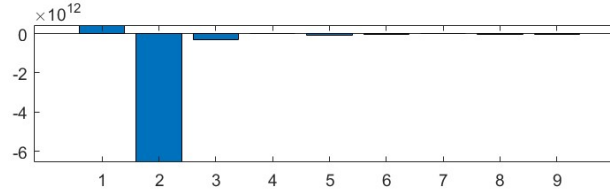
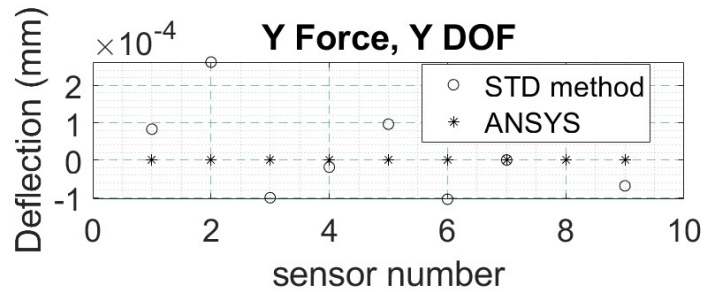
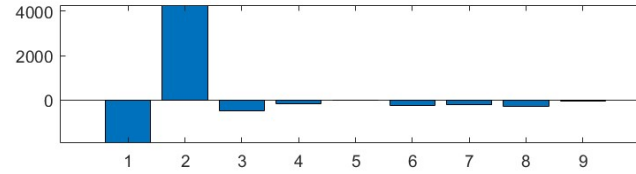
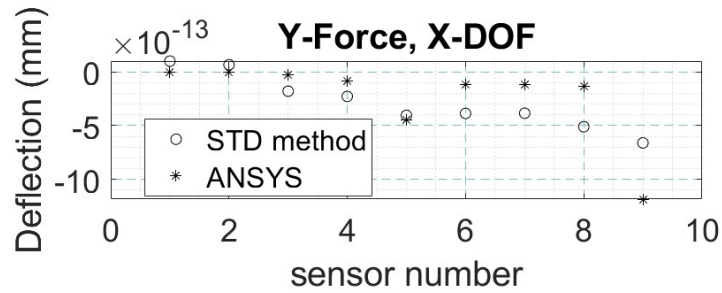
Results – Top Sensors, Y-direction Forcing



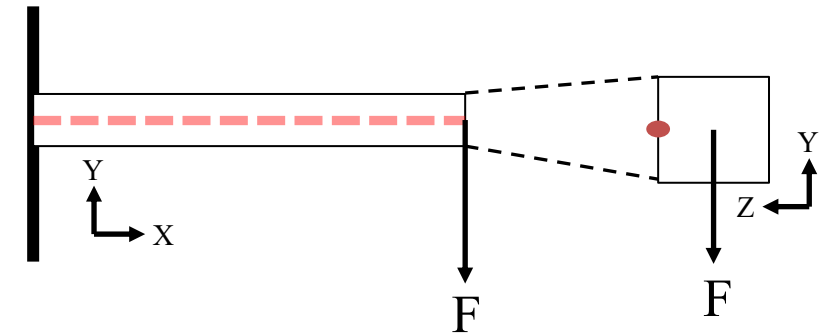
- Results from top sensors and side sensors from a Y-force displayed, as well as top sensors with an X-force, with errors on the right
- Strain prediction through stress readings most successful in case where normal strain in each axis was dominant strain, rather than shear strain
- When opposite case was true, prediction and true value completely conflicted in alignment



Results – Side Sensors, Y-direction Forcing



- Results from top sensors and side sensors from a Y-force displayed, as well as top sensors with an X-force, with errors on the right
- Strain prediction through stress readings most successful in case where normal strain in each axis was dominant strain, rather than shear strain
- When opposite case was true, prediction and true value completely conflicted in alignment



Thank you!

